

Marking of parts

Marking of materials

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Foreword

The German version is binding.

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This Group Standard has been coordinated with the responsible departments of the BMW Group. Author, approver and reviewer are stored in and can be retrieved from the TERE system master data.

The specifications described in this Group Standard ensure compliance with Directive 2000/53/EC on end-of-life vehicles including its appendices with regard to the marking of materials.

This Group Standard is based on VDA 260.

Described are supplementary application rules and rules applicable for marking.

Amendments

The following amendments have been made to GS 91003:2004-02:

- section 3 added;
- section 3.2 updated, is now section 4.2;
- section 3.2.1 updated and section “Explanations” integrated, is now section 4.2.1;
- cited standards updated;
- Group Standard editorially revised.

Previous editions

BMWN(S) 11310.0: 1991-04, 1991-11, 1993-08, 1999-09

BMWN(S) 11310.0 BBL1: 1998-12

GS 91003: 1999-09, 2004-02

1 Scope and purpose

Marking is carried out on vehicle parts which require the material to be identified for recycling and disposal or for processing in repair operations.

Exceptions are:

- standard parts (e.g. fasteners) and parts similar to standard parts;
- parts where, for technical reasons (function, quality, surface) and/or lack of space, marking is not possible for compelling reasons;
- parts or constituents of them, that are very difficult to dismantle or mark, in relation to their weight;
- if the scope of application is differentiated, in case of doubt, the relevant specialist departments for materials shall be consulted.

2 Normative references

This Group Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

GS 91001	Marking of parts; With trademark and part identification data
GS 93016	Plastics; Thermoplastics; Selection list
DIN 60001-1	Textile fibres; Part 1: Natural fibres and letter codes
DIN ISO 1629	Rubber and latices; Nomenclature
DIN EN ISO 2076	Textiles; Man-made fibres; Generic names
DIN EN ISO 1043-1	Plastics; Symbols and abbreviated terms; Part 1: Basic polymers and their special characteristics
DIN EN ISO 1043-2	Plastics; Symbols and abbreviated terms; Part 2: Fillers and reinforcing materials
DIN EN ISO 11469	Plastics; Generic identification and marking of plastics products
VDA 260	Components of motor vehicles; Marking of materials
2000/53/EC ¹⁾	Directive 2000/53/EC of the European Parliament and of the Council of 18 September 2000 on end-of-life vehicles

1) see <http://eur-lex.europa.eu>

3 Definitions

Separability A part can be dismantled by simple manual means and auxiliary means (e.g. screwing, separating) with or without the use of force (e.g. breaking off, pulling-off).

4 Structure

4.1 Material code

The material code is the short designation of the material as defined in DIN, EN and ISO standards, which is delimited by arrow symbols (> <) during marking. For better readability, a space shall be inserted between the arrow symbols and the material designation.

The material marking shall be prescribed in the drawing.

4.2 Polymer-based materials

Marking shall be structured according to DIN EN ISO 11469.

The material code is based on the standardized code letters and codes stated in DIN EN ISO 1043-1 and DIN ISO 1629.

The material designation is made up as follows:

- a) material code;
- b) type (first letter) and form (second letter) of the filler or reinforcement materials (DIN EN ISO 1043-2, DIN 60001-1, DIN EN ISO 2076) as a mass percentage. Multiple fillers shall be stated in round brackets.

The material code shall be composed according to the descriptions and regulations listed above.

NOTE For the materials released by BMW, material designations are usually documented in internal material standards, e.g. GS 93016.

If the material contains several fillers and/or reinforcing agents, these are separated by a “+” sign (placed in round brackets) in an order corresponding to their proportions by weight (with the highest value at the left), followed by their total proportion as a mass percentage.

If there are more than 3 different fillers and/or reinforcing agents (e.g. in the case of non-standard fiber blend), the abbreviation “XX” is used to represent one or more specific designations, followed by the total proportion as a mass percentage.

A single “X” can be used to represent unspecified or unknown forms of filler or reinforcing agent.

The specifications listed in a) and b) shall be separated using a dash. There is no need to specify b) for elastomer parts.

For materials with recycled material content, the percentage of the recycled material is stated in the material code, e.g. "(REC20)" for a recycled material content of 20 %. If the part is composed completely of recycled material, then it is marked "(REC100)" accordingly.

Table 1: Examples

Material code	Explanation
> ABS + PC-GF15 <	Poly blend of acrylonitrile butadiene styrene and polycarbonate with a mass percentage of 15 % Glass fibers
> PA66-(GF+GB)30 <	Polyamide 66 with glass fibers and glass spheres, Overall proportion, mass percentage of 30 %
> PA66-(MX+GF)30 <	Polyamide 66 with mineral filler (form unknown) and glass fibers, Overall proportion, mass percentage of 30 %
> UP-XX40 <	Unsaturated polyester resin with a fiber blend (e.g. wood, paper, glass, textile), Overall proportion, mass percentage 40 %
> EP-(GF+TV+XX)45 <	Epoxy resin with glass fiber (e.g. 25 %) and talc (e.g. 15 %) additives as the main blend elements and small amounts of other admixtures (e.g. 3 % wood flour plus 2 % paper shavings)
> SBR+NR <	Vulcanized blend, basis styrene-butadiene and natural rubber
> PP-GF40 (REC20) <	Polypropylene homopolymer with overall proportion, mass percentage 40 % glass fibers and a recycled material proportion of 20 %

4.2.1 Position specifications

If the part consists of several separable material constituents or individual parts (e.g. composites, assembly parts) and is externally recognizable as such, these shall preferably be marked individually. In case of space or comprehension problems, the material data shall be attached together in a suitable place (see GS 91001).

For an unambiguous reference between the materials and the position on the part, the individual material designations shall be preceded by a position specification. These position specifications consist of simple words such as top, bottom, left, right, inside, middle, outside, etc. and are formulated in English for international understanding. The position specification always relates to the orientation (reading direction) of the marking.

Specific terminology or part names (e.g. "top part", "carrier") shall only be used in exceptional circumstances for clearer understanding.

It is also possible to use position symbols or pictograms, but these shall be clear and self-explanatory.

EXAMPLE Hard-soft (rigid-resilient) composite material

IN	> PA6-GF30 <
OUT	> PUR-T <

The design of position specifications and part numbers is based on the premise of practicality and comprehensibility with regard to a homogeneous dismantling and separation process. The designation is determined by Engineering specifically for the part on the basis of the options listed.

In individual cases, it may be necessary or beneficial to create mixed forms, but these shall be approved by the Plastics Development specialist department, Materials functional area.

If the constituents integrated in the part are joined together in such a way that it is not possible or reasonable to separate them homogeneously by type, the position specification may be omitted following consultation with the above-mentioned specialist departments.

In the case of molded parts that comprise two or more plastic components that are not externally visible, shall be marked such that the material visible on the surface is marked first, followed by the other materials separated by commas.

EXAMPLE Layer structure of a co-extruded tube:

> PA11, PA6, PA11 <

The material structure is clearly defined here even without a position specification. Furthermore, separating the components can be viewed as problematic.

4.3 Metals, metallic coatings

Only parts made of materials whose homogeneous recovery is economically reasonable or necessary for environmental protection reasons, are marked.

The code for the material according to the respective material standards is used for marking. The code is also placed between the arrow symbols as described in section 4.1.

Coatings that either cause recycling problems or whose recovery is reasonable shall be identified by indicating the basic material, followed by a forward and the coating material.

Table 2: Examples

Material code	Explanation
> X5CrNi18-10 <	high-alloy steel with $\leq 0,07$ % C, (17,5 to 19,5) % Cr, (8,0 to 10,5) % Ni, small quantities of further constituents
> EN-GJS <	cast iron with spheroidal graphite (formerly GGG)
> EN-MCMgAl8Zn1 <	Magnesium alloy with (7,0 to 8,7) % Al, (0,35 to 1,0) % Zn, small quantities of further constituents, remainder Mg
> EN AW-ALSi10Mg1,5 <	Aluminum alloy with (9,0 to 10,5) % Si, (1 to 2) % Mg, small quantities of further constituents, remainder Al
> Fe/Zn <	zinc coated sheet steel

5 Design

Positioning and design of the marking shall be according to GS 91001.

In the case of geometrically long, thin-walled or large-area parts (strips, bumpers, etc.), the marking shall be applied several times so that sections can still be identified.

6 Acceptance

The design of the initial sample is decisive for the acceptance. Any alteration shall be initiated by a new sampling inspection and requires a renewed release in written form.